

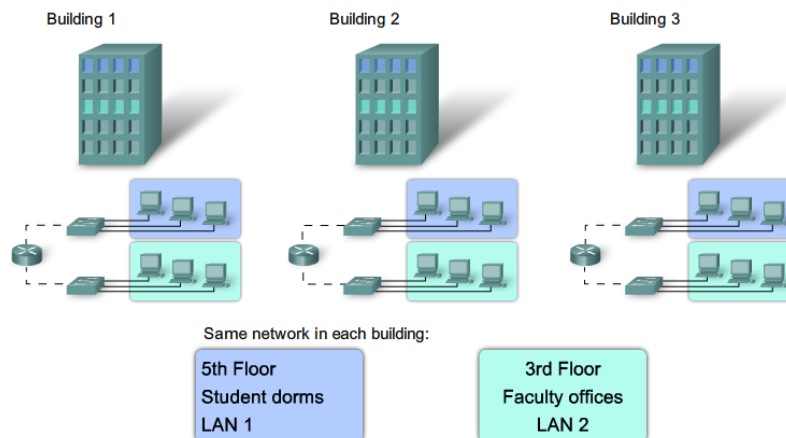
The University of Jordan (UJ)
School of Engineering
Department of Computer Engineering
Advanced Networks Lab 0907529
Exp.1 Virtual Local Area Networks (VLANs)

Objectives

1. Explain the role of VLANs in a network.
2. Explain the role of trunking in a network.
3. Configure VLANs and trunks on switches.
4. Verify VLAN Information

Overview of VLANs

To appreciate why VLANs are being widely used today, consider a small community college with student dorms and the faculty offices all in one building (Building 1). The figure below shows the student computers in one LAN and the faculty computers in another LAN. This works fine because each department is physically together, so it is easy to provide them with their network resources. A year later, the college has grown and now has three buildings. In the figure, the original network is the same, but student and faculty computers are spread out across three buildings. The student dorms remain on the fifth floor and the faculty offices remain on the third floor. However, now the IT department wants to ensure that student computers all share the same security features and bandwidth controls. How can the network accommodate the shared needs of the geographically separated departments? Do you create a large LAN and wire each department together? How easy would it be to make changes to that network? It would be great to group the people with the resources they use regardless of their geographic location, and it would make it easier to manage their specific security and bandwidth needs.



The solution for the community college is to use a networking technology called a virtual LAN (VLAN). A VLAN allows a network administrator to create groups of logically networked devices that act as if they are on their own independent network, even if they share a common infrastructure with other VLANs. When you configure a VLAN, you can name it to describe the primary role of the users for that VLAN. As an example, all of the student computers in a school can be configured in the "Student" VLAN. Using VLANs, you can logically segment switched networks based on functions, departments, or project teams. These VLANs allow the network administrator to implement access and security policies to particular groups of users.

What are VLANs?

A VLAN (Virtual Local Area Network) is a logical network that groups devices regardless of their physical location. Each VLAN acts as a separate subnet. Access ports connect to one VLAN, while routers are needed for communication between VLANs.

Benefits of a VLAN Design

- Security: Isolate sensitive data (e.g., faculty vs students).
- Cost reduction: Better use of existing infrastructure.
- Performance: Reduce broadcast traffic and increase efficiency.
- Broadcast storm mitigation: Limit propagation across the network.
- Easier management: VLANs can be clearly named (e.g., Student, Faculty, Guest).
- Simplified project/application management: Easier to manage specialized teams.

VLAN ID Ranges

Access VLANs are divided into either a normal range or an extended range.

1. Normal Range VLANs
 - Used in small- and medium-sized business and enterprise networks.
 - Identified by a VLAN ID between 1 and 1005.
 - IDs 1002 through 1005 are reserved for Token Ring and FDDI VLANs.
 - IDs 1 and 1002 to 1005 are automatically created and cannot be removed.
 - Configurations are stored within a VLAN database file, called vlan.dat. The vlan.dat file is located in the flash memory of the switch.
2. Extended Range VLANs
 - Enable service providers to extend their infrastructure to a greater number of customers. Some global enterprises could be large enough to need extended range VLAN IDs.
 - Are identified by a VLAN ID between 1006 and 4094.
 - Support fewer VLAN features than normal range VLANs.
 - Are saved in the running configuration file.

Types of VLANs

- 1- Default VLAN
 - All switch ports are automatically placed in VLAN 1.
 - VLAN 1 is the **default native VLAN**.
 - VLAN 1 is also the **default management VLAN**.
 - VLAN 1 **cannot be renamed or deleted**.
- 2- Data VLAN
 - Used to carry **user-generated traffic** only.
 - Also called a **user VLAN** (groups users/devices).
 - Modern networks often use **multiple data VLANs**.
 - **Voice and management traffic are not allowed** on data VLANs.

3- Native VLAN

- Traffic from a VLAN must carry its VLAN ID when sent to another switch.
- **Trunk ports** connect switches and carry tagged traffic.
- **802.1Q trunks** add a 4-byte tag to Ethernet frames to indicate the VLAN.
- **Untagged traffic** can be sent over trunks, usually placed on the **native VLAN**.
- On Cisco switches, the default native VLAN is **VLAN 1**.
- Best practice: assign the native VLAN to an **unused VLAN**, different from VLAN 1 and other active VLANs.

4- Management VLAN

- A **management VLAN** handles network management traffic (SSH, Telnet, HTTPS, HTTP, SNMP).
- By default, **VLAN 1** is used for this on a Layer 2 switch.

5- Voice VLANs

A **Voice VLAN** is used to support VoIP traffic, ensuring low delay and priority to maintain call quality. The details of how to configure a network to support VoIP are beyond the scope of this experiment.

Defining VLAN Trunks

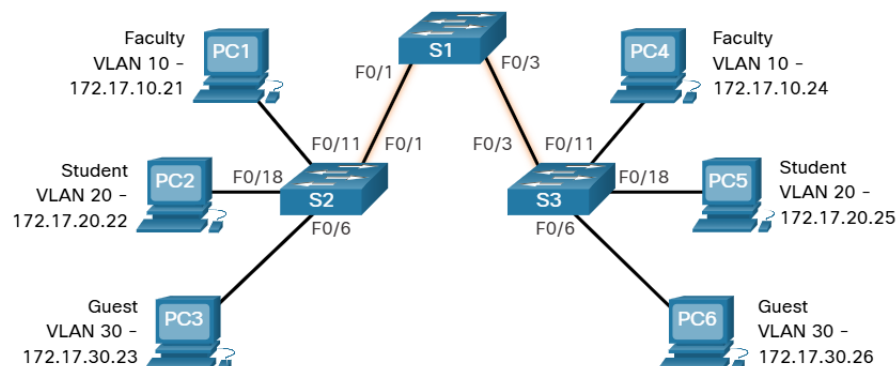
What is a trunk?

- A trunk is a connection (link) between two network devices (usually switches).
- It allows **multiple VLANs** to pass through the same cable.
- This makes it possible for devices on the same VLAN but on different switches to communicate **without a router**.

Trunk features:

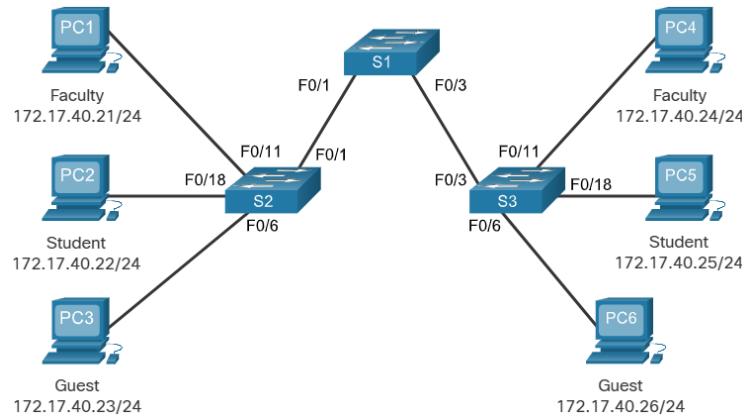
- Uses the **IEEE 802.1Q** standard to add a tag to each frame, showing which VLAN it belongs to.
- A trunk is not tied to one VLAN; it carries all the allowed VLANs.
- Can connect not only switches but also servers or devices with a network card that supports 802.1Q.
- By default, Cisco switches allow all VLANs on a trunk port.

In the figure below, the highlighted links between switches S1 and S2, and S1 and S3 are configured to transmit traffic coming from VLANs 10, 20, 30, and 99 (i.e., native VLAN) across the network. This network could not function without VLAN trunks.



Network without VLANs

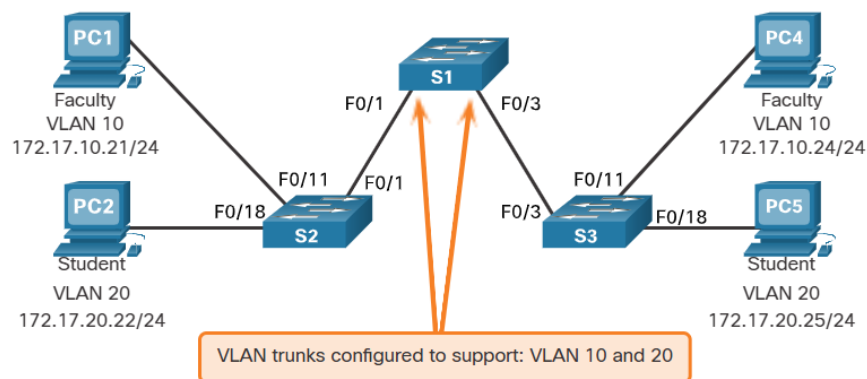
When a switch receives a broadcast frame on one of its ports, it forwards the frame out all other ports except the port where the broadcast was received. In the figure, the entire network is configured in the same subnet (172.17.40.0/24) and no VLANs are configured. As a result, when the faculty computer (PC1) sends out a broadcast frame, switch S2 sends that broadcast frame out all of its ports. Eventually the entire network receives the broadcast because the network is one broadcast domain.



Network with VLANs

VLANs are associated with and configured on individual switch ports. Devices attached to those ports have no concept of VLANs. However, these devices are configured with IP addressing and are members of a specific IP network. This is where the connection between VLAN and IP network is apparent. A VLAN is the equivalent to an IP network (or subnet). VLANs are configured on the switch, whereas IP addressing is configured on the device.

In the figure, the same network has now been segmented using two VLANs. Faculty devices are assigned to VLAN 10 and student devices are assigned to VLAN 20. When a broadcast frame is sent from the faculty computer, PC1, to switch S2, the switch forwards that broadcast frame only to those switch ports configured to support VLAN 10.



The ports that comprise the connection between switches S2 and S1 (ports F0/1), and between S1 and S3 (ports F0/3) are trunks and have been configured to support all the VLANs in the network.

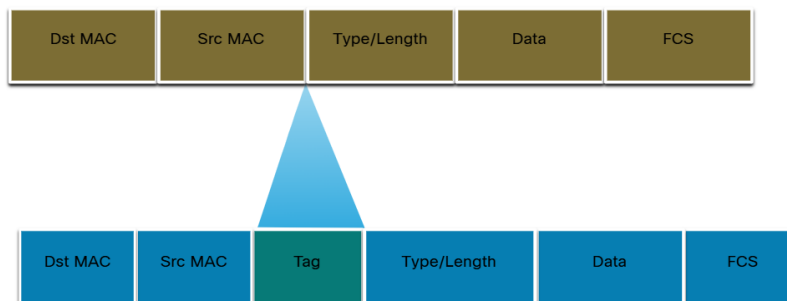
When S1 receives the broadcast frame on port F0/1, S1 forwards that broadcast frame out of the only other port configured to support VLAN 10, which is port F0/3. When S3 receives the broadcast frame on port F0/3, it forwards that broadcast frame out the only other port configured to support

VLAN 10, which is port F0/11. The broadcast frame arrives at the only other computer in the network configured in VLAN 10, which is faculty computer PC4.

VLAN Identification with a Tag

The standard Ethernet frame header does not contain information about the VLAN to which the frame belongs. Therefore, when Ethernet frames are placed on a trunk, information about the VLANs to which they belong must be added. This process, called tagging, is accomplished by using the IEEE 802.1Q header, specified in the IEEE 802.1Q standard. The 802.1Q header includes a 4-byte tag inserted within the original Ethernet frame header, specifying the VLAN to which the frame belongs.

When the switch receives a frame on a port configured in access mode and assigned a VLAN, the switch inserts a VLAN tag in the frame header, recalculates the Frame Check Sequence (FCS), and sends the tagged frame out of a trunk port.



VLAN Creation Commands

When configuring normal range VLANs, the configuration details are stored in flash memory on the switch in a file called vlan.dat. Flash memory is persistent and does not require the **copy running-config startup-config** command. However, because other details are often configured on a Cisco switch at the same time that VLANs are created, it is good practice to save running configuration changes to the startup configuration.

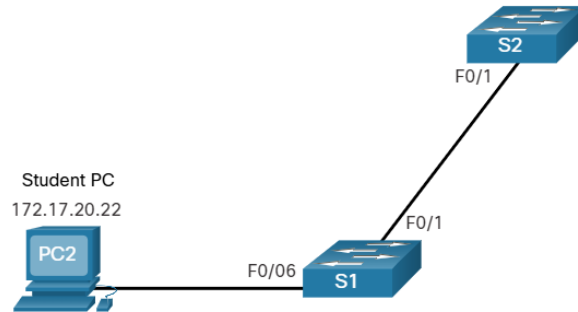
The table displays the Cisco IOS command syntax used to add a VLAN to a switch and give it a name. Naming each VLAN is considered a best practice in switch configuration.

* Create vlans

Task	IOS Command
Enter global configuration mode.	<code>Switch# configure terminal</code>
Create a VLAN with a valid ID number.	<code>Switch(config)# vlan vlan-id</code>
Specify a unique name to identify the VLAN.	<code>Switch(config-vlan)# name vlan-name</code>
Return to the privileged EXEC mode.	<code>Switch(config-vlan)# end</code>

sh vlan
default → 1
1002 ... 1005
5 vlans
[2-1001]
do

In the topology, the student computer (PC2) has not been associated with a VLAN yet, but it does have an IP address of 172.17.20.22, which belongs to VLAN 20.



The example shows how the student VLAN (VLAN 20) is configured on switch S1.

```

S1# configure terminal
S1(config)# vlan 20
S1(config-vlan)# name student
S1(config-vlan)# end
  
```

VLAN Port Assignment Commands

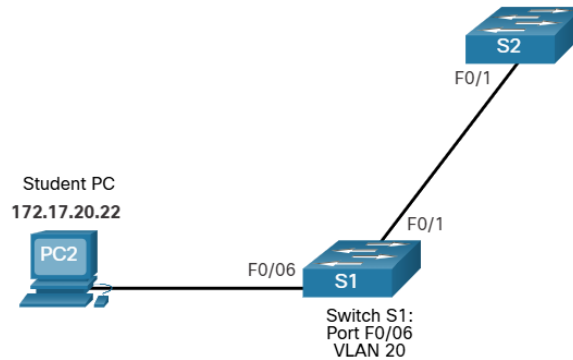
After creating a VLAN, the next step is to assign ports to the VLAN. The table displays the syntax for defining a port to be an access port and assigning it to a VLAN. The switchport mode access command is optional, but strongly recommended as a security best practice. With this command, the interface changes to strictly access mode. Access mode indicates that the port belongs to a single VLAN and will not negotiate to become a trunk link.

** Give Interface*

Task	IOS Command
Enter global configuration mode.	Switch# <u>configure terminal</u>
Enter interface configuration mode.	Switch(config)# <u>interface interface-id</u>
Set the port to access mode.	Switch(config-if)# <u>switchport mode access</u>
Assign the port to a VLAN.	Switch(config-if)# <u>switchport access vlan vlan-id</u>
Return to the privileged EXEC mode.	Switch(config-if)# <u>end</u>

Note: Use the **interface range** command to simultaneously configure multiple interfaces.

In the figure, port F0/6 on switch S1 is configured as an access port and assigned to VLAN 20. Any device connected to that port will be associated with VLAN 20. Therefore, in our example, PC2 is in VLAN 20.



In the figure above, port F0/6 on switch S1 is configured as an access port and assigned to VLAN 20. Any device connected to that port will be associated with VLAN 20. Therefore, in our example, PC2 is in VLAN 20.

```

S1# configure terminal
S1(config)# interface fa0/6
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 20
S1(config-if)# end
  
```

VLANs are configured on the switch port and not on the end device. PC2 is configured with an IPv4 address and subnet mask that is associated with the VLAN, which is configured on the switch port. In this example, it is VLAN 20. When VLAN 20 is configured on other switches, the network administrator must configure the other student computers to be in the same subnet as PC2 (172.17.20.0/24).

Verify VLAN Information

After a VLAN is configured, VLAN configurations can be validated using Cisco IOS **show** commands.

The **show vlan** command displays a list of all configured VLANs. The **show vlan** command can also be used with options. The complete syntax is **show vlan [brief | id vlan-id | name vlan-name | summary]**.

** shows*

The table describes the show vlan command options.

Task	Command Option
Display VLAN name, status, and its ports one VLAN per line.	brief
Display information about the identified VLAN ID number. For <i>vlan-id</i> , the range is 1 to 4094.	id vlan-id
Display information about the identified VLAN name. The <i>vlan-name</i> is an ASCII string from 1 to 32 characters.	name vlan-name
Display VLAN summary information.	summary

The show vlan summary command displays the count of all configured VLANs.

```
S1# show vlan summary
Number of existing VLANs      : 7
Number of existing VTP VLANs  : 7
Number of existing extended VLANs : 0
```

Other useful commands are the **show interfaces interface-id switchport** and the **show interfaces vlan vlan-id** command. For example, the **show interfaces fa0/18 switchport** command can be used to confirm that the FastEthernet 0/18 port has been correctly assigned to data and voice VLANs.

```
S1# show interfaces fa0/18 switchport
Name: Fa0/18
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 20 (student)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: 150
Administrative private-vlan host-association: none
(Output omitted)
```

Change VLAN Port Membership

If the switch access port has been incorrectly assigned to a VLAN, then simply re-enter the **switchport access vlan vlan-id** interface configuration command with the correct VLAN ID. For instance, assume Fa0/18 was incorrectly configured to be on the default VLAN 1 instead of VLAN 20. To change the port to VLAN 20, simply enter **switchport access vlan 20**.

To change the membership of a port back to the default VLAN 1, use the **no switchport access vlan** interface configuration mode command as shown.

In the output for example, Fa0/18 is configured to be on the default VLAN 1 as confirmed by the **show vlan brief** command.

```
S1(config)# interface fa0/18
S1(config-if)# no switchport access vlan
S1(config-if)# end
S1#
S1# show vlan brief
VLAN Name      Status    Ports
-----
1  default      active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                Gi0/1, Gi0/2
20  student      active
1002 fddi-default  act/unsup
1003 token-ring-default act/unsup
1004 fddinet-default  act/unsup
1005 trnet-default  act/unsup
```

Notice that VLAN 20 is still active, even though no ports are assigned to it.

*Remote access on switch:-

```
vlan 99
name mgt
exit
enable secret cisco
line vty 0 15
password vlypass
login
exit
interface F0/20
switchport mode access
switchport access vlan 99
interface vlan 99
ip address 172.168.99.1
255.255.255.0
```

switch

give Ip on same Network
try telnet 172.168.99.1

جو یہ راستے پر PC تبصرے
switch پر

PC

Delete VLANs

The **no vlan *vlan-id*** global configuration mode command is used to remove a VLAN from the switch `vlan.dat` file.

Caution: Before deleting a VLAN, reassign all member ports to a different VLAN first. Any ports that are not moved to an active VLAN are unable to communicate with other hosts after the VLAN is deleted and until they are assigned to an active VLAN.

The entire `vlan.dat` file can be deleted using the **delete flash:vlan.dat** privileged EXEC mode command. The abbreviated command version (**delete vlan.dat**) can be used if the `vlan.dat` file has not been moved from its default location. After issuing this command and reloading the switch, any previously configured VLANs are no longer present.

Trunk Configuration Commands

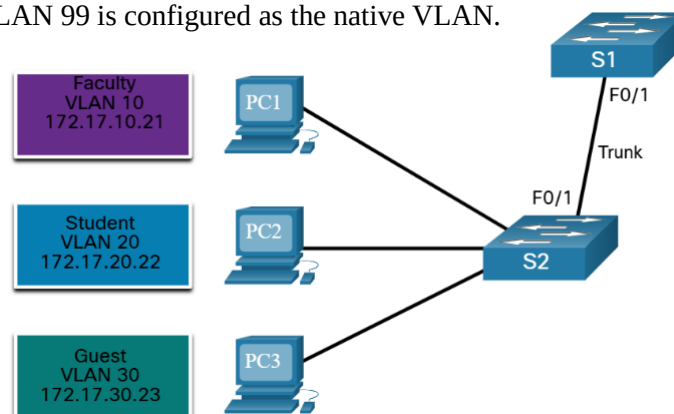
**trunk*

Now that you have configured and verified VLANs, it is time to configure and verify VLAN trunks. A VLAN trunk is a Layer 2 link between two switches that carries traffic for all VLANs (unless the allowed VLAN list is restricted manually or dynamically). To enable trunk links, configure the interconnecting ports with the set of interface configuration commands shown in the table.

Task	IOS Command
Enter global configuration mode.	<code>Switch# configure terminal</code>
Enter interface configuration mode.	<code>Switch(config)# interface interface-id</code>
Set the port to permanent trunking mode.	<code>Switch(config-if)# switchport mode trunk</code>
Sets the native VLAN to something other than VLAN 1.	<code>Switch(config-if)# switchport trunk native vlan <i>vlan-id</i></code>
Specify the list of VLANs to be allowed on the trunk link.	<code>Switch(config-if)# switchport trunk allowed vlan <i>vlan-list</i></code>
Return to the privileged EXEC mode.	<code>Switch(config-if)# end</code>

*if untagged traffic
it goes to the
native
vlan
1,50,99,...*

In the figure, VLANs 10, 20, and 30 support the Faculty, Student, and Guest computers (PC1, PC2, and PC3). The F0/1 port on switch S1 is configured as a trunk port and forwards traffic for VLANs 10, 20, and 30. VLAN 99 is configured as the native VLAN.



The example shows the configuration of port F0/1 on switch S1 as a trunk port. The native VLAN is changed to VLAN 99 and the allowed VLAN list is restricted to 10, 20, 30, and 99.

```
S1(config)# interface fastEthernet 0/1
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk native vlan 99
S1(config-if)# switchport trunk allowed vlan 10,20,30,99
S1(config-if)# end
```

Verify Trunk Configuration

The switch output displays the configuration of switch port F0/1 on switch S1. The configuration is verified with the **show interfaces interface-ID switchport** command.

```
S1# show interfaces fa0/1 switchport
Name: Fa0/1
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 99 (VLAN0099)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk associations: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: 10,20,30,99
Pruning VLANs Enabled: 2-1001
(output omitted)
```

The top highlighted area shows that port F0/1 has its administrative mode set to **trunk**. The port is in trunking mode. The next highlighted area verifies that the native VLAN is VLAN 99. Further down in the output, the bottom highlighted area shows that VLANs 10, 20, 30, and 99 are enabled on the trunk. **Note:** Another useful command for verifying trunk interfaces is the **show interface trunk** command.

Procedures:

Dear students, please note that the lab problems sheet, the packet tracer activities and the practical discussion videos have been uploaded on your Microsoft Teams group. You are required to carefully study this experiment and then complete the lab sheet.

References

Cisco Networking Academy - CCNA: Switching, Routing, and Wireless Essentials.
<https://www.netacad.com>